

When Does BERT Help?

A Cross-Domain Study of the Marginal Value of Text Embeddings Beyond Structured Features

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Abstract—Adding a pretrained text embedding to a tabular model is practically a standard step in applied machine learning. However, we rarely ask whether the text embedding provides additional information beyond structured and metadata features. That is what we do in this study: We combine frozen DistilBERT embeddings, using the masked mean pooling technique, and domain-specific structured features and train Random Forest classifiers in four separate setups: popularity classification and topic classification for news articles and Reddit posts. For each setup we compute SHAP feature-group importance as well as group-level permutation importance as an additional robustness check. In addition, we compute paired-bootstrap 95% confidence intervals (with 2,000 bootstrap resamples) for the difference in accuracy between tabular-only and tabular+BERT setups. The findings are quite specific: the two importance criteria agree on which structured feature group matters most in every setup. The BERT model brings statistically significant gain in only one of the four setups: for Reddit topic (subreddit) classification, in which case the accuracy increases by 17.8 percentage points (p less than 0.001). As for Reddit popularity prediction, the addition of BERT makes the classifier accuracy decrease by 2.9 points (p equals 0.008). Neither of the two news domain tasks shows any statistically significant change in accuracy (p between 0.06 and 0.08). The structural and metadata features remain the major contributors to per-feature importance in three out of four setups, regardless of adding BERT. The findings suggest an important boundary condition for frozen text embeddings. The additional dimensionality of their representation turns out to be justified only in case of lexical or topical target prediction tasks. Behavioral targets, or targets that are well-represented through structured features, are unlikely to benefit from adding BERT.

Index Terms—feature importance, SHAP, permutation importance, BERT, multimodal learning, popularity prediction, topic classification, bootstrap confidence intervals

I. INTRODUCTION

Practitioners routinely bolt a pretrained text embedding onto an existing tabular pipeline under the assumption that “more signal can’t hurt.” However, it becomes significant to evaluate this assumption against the null hypothesis that the added usefulness of an embedding depends strongly on the type of target being predicted. Text embeddings are more likely to help when the target is closely tied to lexical or topical information than when it is primarily determined by behavioral or structural characteristics.

The Online News Popularity dataset [1] provides a useful setting for examining this distinction. Earlier studies of this dataset, as well as broader research on social-content popularity [2]–[4], have consistently found that structural and metadata variables—such as publication timing, keyword statistics, link counts, and self-reference measures—carry substantial predictive power, while the article text itself contributes comparatively little. The more precise question, then, is how much information the text embedding actually adds once the large dimensionality gap between a 768-dimensional embedding representation and a small number of engineered features is taken into account. It is also worth asking whether this relationship is unique to news popularity or extends to other datasets and prediction tasks.

We address this with a controlled 2×2 study:

- Two domains capturing contrasting ends of the same content spectrum: professionally produced news articles written in a conventional journalistic style, represented by structural and metadata variables, sentiment and topic-model features from the UCI Online News Popularity dataset, and re-scraped article text; and informal, user-generated Reddit posts, represented by structural and metadata variables together with self-post titles and text from a balanced sample spanning 16 subreddits.
- Two tasks for each domain: popularity classification, based on quartile bins of shares or score, and topic classification, based on news channel or subreddit identity. The topic task serves as a positive control because the lexical properties of an article or post are inherently connected to its label.
- Two feature configurations: a tabular-only model and a combined tabular-plus-text model using frozen DistilBERT embeddings of titles and text with masked mean pooling. This setup follows the multimodal tabular-text modeling approach adopted by AutoGluon [5].

For all eight domain $\times$ task $\times$ configuration permutations, we present three complementary results:

- SHAP feature-group importance scores [6] are computed for the raw group-level sums as well as normalised per-

feature. Normalisation is required due to the large size (dimensionality = 768) of the BERT features.

- Group-level permutation-based importances are calculated [7], [8] to check if the discovered patterns are unique to TreeExplainer.
- Accuracy difference between the tabular-only and tabular+BERT models is tested for statistical significance via a paired bootstrap method. The purpose is to validate our claims about either improvement or deterioration of the BERT model performance based on confidence intervals.

Contributions

- We provide a reusable validation technique based on SHAP-normalized values, a permutation-importance test, and bootstrapped confidence intervals. The combination of techniques examines whether the inclusion of an additional feature component based on the high-dimensional embedding space adds significant value to the low-dimensional, structured input features.
- We establish an empirical threshold condition that holds consistently true in two different settings of the attribution task. The frozen BERT model makes a statistically significant positive contribution only when the task involves a label related to topics (e.g., classification of subreddits). In the other three domain-task combinations, the contribution is either statistically insignificant or statistically negative.
- We provide a fully reproducible pipeline including data loading, embeddings, classification, and attribution stages for both datasets (code and exact software/hardware details in §IV-D).

Multimodality as a method of text and table fusion is anything but new [5]. Yet, what we offer is a method of statistical analysis of when such a known approach brings valuable results, and when it fails.

II. RELATED WORK

Popularity prediction on the UCI Online News Popularity dataset. Fernandes et al. [1] released the dataset and formulated it as intelligent decision support; raw text of the articles is not included in the original UCI dataset, only 58 numeric engineered features (sentiment, LDA topic scores, keyword counts etc.). Since the original webpages are not part of the dataset, testing the value of the text embedding requires us to scrape them again, and we do that for a sub-set of 3,437 articles (§III).

Content-driven vs. structural predictors of popularity. One common theme from studies of social media virality is that structural and context variables (posting time, author/history of posting venue, network position) account for most of the variation in popularity, whereas content-specific variables do not explain much after controlling for those confounding factors [2]–[4]. Our results on news/popularity and Reddit/popularity are thus consistent with and expand upon this line of research using the SHAP/permutation framework with a state-of-the-art transformer embedding of the text data.

Multimodal tabular+text fusion. From the AutoGluon multimodal AutoML benchmark [5], we can conclude that the established and validated way of combining tabular features with text embeddings is the late/early fusion approach. We choose the simplest implementation of fusion — embedding concatenation with the tabular model — to ensure that there is no way to attribute a negative outcome to a poor choice of fusion. Bonnier [9] advances the research in this direction by suggesting a two-stream tabular-text transformer based on cross-modal attention, which the author claims to be better than mere concatenation of the features, while Xu et al. [10] review a variety of multimodal fusion approaches using transformers. This is exactly why we do not use the more expressive multimodal fusion architectures, because our aim is to investigate whether the static embedding carries any information at all.

Methods for attribution. SHAP [6] offers theoretically well-founded per-feature attributions for tree ensembles by means of TreeExplainer; permutation importance [7], [8] is an agnostic method to the model and addresses a slightly different question (how much accuracy decreases when features are shuffled, as opposed to Shapley-value attributions). More recent work has continued to refine Shapley-based attribution itself: Kwon and Zou [11] show that the uniform weighting in standard Shapley values can be suboptimal for identifying influential features and propose a learned reweighting. A separate, model-agnostic family of explanation methods, exemplified by LIME [12], instead fits a local surrogate model around each prediction rather than computing exact Shapley values; we do not use LIME here since it lacks TreeExplainer’s exactness guarantee for tree ensembles. We use standard (unweighted) SHAP here, cross-checked against permutation importance, since our goal is a robustness check across attribution frameworks rather than the most sample-efficient attribution estimator. Both methods are employed here explicitly, since in principle, they might provide inconsistent results, and consistency is a valuable confirmation of findings.

III. DATA AND TASK DESIGN

The two domains chosen are a conscious choice, rather than a convenience sample. News articles and Reddit posts have been carefully chosen to be at opposite ends of the same register continuum: news articles are polished, professional writing written according to the conventions of newswriting (headline, lead, body, and style of publication), while Reddit posts are unpolished, casual writing, often using slang and abbreviations. Assuming that the usefulness of an out-of-the-box frozen language model is determined primarily by the match between the language model’s pretraining distribution and the register of the target text, we would have expected the performance to be similar for each domain depending on the task. In contrast (§V), BERT’s usefulness differs much more within the domain across the popularity/topic task split, rather than between the two domains — a sign that the source of BERT’s usefulness is the semantic structure of the label, not writing register.

TABLE I
DATASET SUMMARY

	News	Reddit
Popularity task, n	3,437	15,998
Popularity classes	4 (quartile)	4 (quartile)
Popularity chance level	25.0%	25.0%
Topic task, n	2,885 ^a	15,998
Topic classes	6 (channel)	16 (subreddit)
Topic chance level	16.7%	6.25%
Topic majority baseline	25.5%	6.25% ^b
Text embedding dim.	768	768
Structured/meta feature count	43 (37 for topic)	17 (16 for topic)

^aSubset with a single-channel label (one-hot channel indicators sum to exactly 1). ^bReddit sample is near-perfectly balanced across subreddits ($\approx 1,000$ posts each, 16 subreddits).

A. News domain

The base features include the 43 numeric predictors in the UCI Online News Popularity data set [1], which can be split into four groups: sentiment (16 features: subjectivity/polarity statistics for body and title), LDA (5 features: topic-model membership scores), channel (6 one-hot vectors identifying news sections; predictors in the popularity task, and the classification target in the topic task, where they are removed from the feature set), and structural (16 features: statistics on links, images, videos, and keywords). Titles and texts of articles were not provided by the UCI dataset; 3,437 original Mashable URL sources were retrieved directly using the Python newspaper library (live fetching, with a Wayback Machine fallback for dead links) to obtain the text required for the BERT embeddings. The popularity target is the original quartile binning of the `shares` feature into 4 classes. The topic target is the news channel identified through one-hot vectors; rows with zero or multiple active channels ($n = 552$) are removed from the topic task, resulting in 2,885 articles across 6 classes (Table I). Table II lists the exact column-to-group mapping.

B. Reddit domain

The Reddit sample consists of publicly-posted Reddit submissions, accessed under Reddit’s terms of use. A total of 15,998 posts selected randomly and evenly ($\approx 1,000$ each) from 16 subreddits (`AskReddit`, `Jokes`, `tifu`, `explainlikeimfive`, `todayilearned`, `pics`, `aww`, `gifs`, `funny`, `books`, `science`, `technology`, `worldnews`, `sports`, `relationships`, `movies`) to create a close-to-uniform topic prior distribution — an interesting counter-example to news, where such a prior will be inherently unbalanced. Features (17 in total; 16 for the topic prediction task, which excludes `subreddit_code` since it is the label by definition) can be grouped as follows: temporal (hour of the day, day of the week, weekend indicator), source (indicators for the domain type, plus `subreddit_code` for the popularity task), structure (number of tokens in title and selftext, selftext exists, is a self-post, thumbnail image exists, number of comments), and meta (NSFW indicator, presence of question/exclamation marks, number of capitalized words

in the title). The popularity target consists of four within-subreddit score-percentile quartiles, constructed to ensure balanced classes across subreddits (Table I). Table III lists the exact column-to-group mapping.

IV. METHOD

A. Pipeline

Fig. 1 describes the universal pipeline used separately for each dataset. The structured/meta features are extracted from the tabular data source. Missing title or body/selftext fields are treated as empty strings before concatenation; no other text cleaning (e.g., lower-casing, punctuation removal) is applied, since DistilBERT’s own tokenizer handles this. The title and body/selftext are concatenated and sent through a frozen `distilbert-base-uncased` [13] (a distillation of BERT [14], itself built on the Transformer architecture [15]) model, limited to 512 tokens; mean-pooling of token-level hidden states from the final layer using attention masks (masked mean pooling) generates one 768-dimensional vector per document. The weights of the DistilBERT model are not fine-tuned or updated at all during training — the embedding is fixed and works as a feature extractor, decoupling “is there signal in the frozen representation?” from “is there a way to fine-tune this particular language model?”, which is a separate question.

B. Classifier

We adopt a Random Forest classifier [7] (`scikit-learn` [16], $T=200$ trees) instead of a neural fusion model because (1) the former makes the exact SHAP TreeExplainer computation much simpler than the kernel method required in case of neural networks and (2) it is a solid, widely-studied benchmark classifier on a tabular-dominated feature space, so that the contribution of the BERT can only come from the latter. Beyond $T=200$ and the Reddit-only depth cap (§IV-D), all other Random Forest hyperparameters are left at their `scikit-learn` 1.7 defaults (Gini impurity, $\sqrt{n_{\text{features}}}$ per split, bootstrap sampling, no class weighting).

C. Attribution and significance protocol

For each domain-task-configuration cell we calculate the following metrics:

Group SHAP importance. Absolute SHAP values per sample computed with the TreeExplainer are averaged per feature and aggregated per group in two ways: (a) sum of raw values of all features in the group and (b) average per feature (sum of raw values divided by group size). The sums are included for completeness, but are not used in comparison across groups due to mechanical bias towards the group with more features, the 768-dimensional BERT block in our case.

Permutation importance (robustness check). Permuting independently of SHAP all columns of one feature group across rows in the test set (not one at a time to avoid dependence of the computation on the dimensionality of the BERT block), measuring the drop in accuracy, and repeating

TABLE II
NEWS DOMAIN: FEATURE GROUPS AND CONSTITUENT COLUMNS (43 FEATURES TOTAL)

Group	n	Features
Sentiment	16	global_subjectivity, global_sentiment_polarity, global_rate_positive_words, global_rate_negative_words, rate_positive_words, rate_negative_words, avg_positive_polarity, min_positive_polarity, max_positive_polarity, avg_negative_polarity, min_negative_polarity, max_negative_polarity, title_subjectivity, title_sentiment_polarity, abs_title_subjectivity, abs_title_sentiment_polarity
LDA	5	LDA_00, LDA_01, LDA_02, LDA_03, LDA_04
Channel	6	data_channel_is_lifestyle, data_channel_is_bus, data_channel_is_entertainment, data_channel_is_socmed, data_channel_is_tech, data_channel_is_world
Structural	16	num_hrefs, num_self_hrefs, num_imgs, num_videos, kw_min_min, kw_max_min, kw_avg_min, kw_min_max, kw_max_max, kw_avg_max, kw_min_avg, kw_max_avg, kw_avg_avg, self_reference_min_shares, self_reference_max_shares, self_reference_avg_shares

TABLE III
REDDIT DOMAIN: FEATURE GROUPS AND CONSTITUENT COLUMNS (17 FOR POPULARITY, 16 FOR TOPIC)

Group	n	Features
Temporal	3	hour_of_day, day_of_week, is_weekend
Source	4/3 ^a	domain_is_self, domain_is_imgur, domain_is_youtube, subreddit_code ^a
Structure	6	n_tokens_title, n_tokens_selftext, has_selftext, is_self, has_thumbnail, num_comments
Meta	4	over_18, title_has_question, title_has_exclaim, title_num_caps_words

^asubreddit_code is included in the *source* group for the popularity task only (4 features); it is dropped for the topic task (3 features) since it is nearly the label itself.

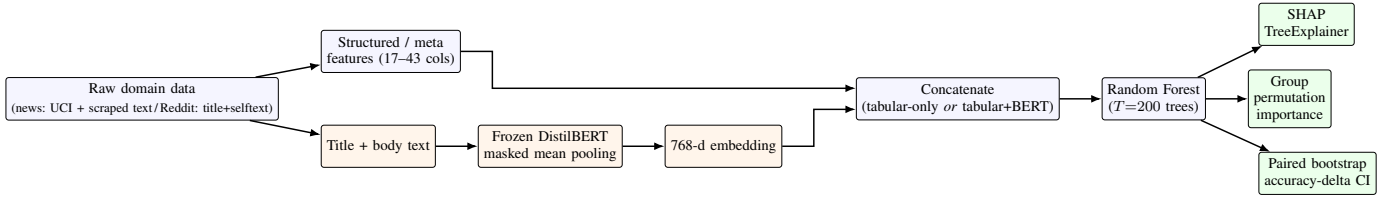


Fig. 1. Shared pipeline. Both feature configurations (tabular-only, tabular+BERT) and all four task/domain cells run this exact pipeline; only the input feature set, the label, and the classifier’s Random Forest hyperparameters (depth cap, Reddit only, §IV-D) differ.

it multiple times ($n_{\text{repeats}} \in \{5, 10\}$ based on dataset size), we obtain the group accuracy drop. It is a different computation from SHAP (drop in accuracy when shuffled vs. Shapley value) while both are using the same trained model, hence their agreement cannot be guaranteed a priori.

Paired bootstrap significance test [17]. The tabular-only and tabular+BERT models are fitted on the same stratified train/test split (exact same seed, exact same row order, hence the per-row predictions are paired) for each task/domain, and then the test-row indices are resampled with replacement $n=2000$ times, to compute their accuracies on each sample and to get an empirical distribution of the difference in accuracy $\Delta = \text{acc}_{\text{tab+BERT}} - \text{acc}_{\text{tab}}$. We compute the 95% percentile confidence interval and a two-tailed bootstrap p -value ($2 \times$ tail probability beyond zero).

D. Experimental setup

All splits are 80/20 stratified with the same fixed seed (42) in all scripts in the pipeline. For Reddit, the tree depth is constrained to be at most 18 ($\text{max_depth} = 18$). This is necessary purely due to the computational limitations of exact TreeExplainer on 15,998 rows $\times$ up to 785 features

and does not change any group ranking: on a stratified 1,000-post subset, capped and uncapped runs give identical group rankings in all four Reddit cells. There are no constraints on the depth of the news-domain trees (since the dataset is small enough, 3,437 rows, and exact TreeExplainer works fast regardless of the depth). Note that Random Forest feature subsampling within splits makes the final accuracy slightly sensitive to column ordering ($\lesssim 2$ points) even for a fixed seed; it does not affect any group importance ranking, which is the same under SHAP and permutation importance regardless of column ordering.

Software and hardware. All experiments use Python 3.10, scikit-learn 1.7 (RandomForestClassifier, train_test_split), PyTorch [18] 2.14 and Hugging Face Transformers [19] 5.16 (for the frozen distilbert-base-uncased forward pass), and the shap package 0.49 (TreeExplainer) for SHAP values. DistilBERT inference runs on an Apple M4 GPU via PyTorch’s Metal Performance Shaders (MPS) backend, 16GB unified memory; Random Forest fitting, SHAP, and permutation importance run on CPU. The full pipeline

TABLE IV

MAIN RESULTS: TABULAR-ONLY VS. TABULAR+BERT ACCURACY, WITH PAIRED-BOOTSTRAP 95% CI ON THE DELTA ($n=2000$ RESAMPLES)

Domain	Task	Tab.	+BERT	Δ [95% CI]
News	Popularity	36.3%	32.1%	-4.2 [-8.6, +0.3]
News	Topic	79.9%	77.8%	-2.1 [-4.3, +0.2]
Reddit	Popularity	33.6%	30.8%	-2.9 [-5.1, -0.7]*
Reddit	Topic	64.7%	82.5%	+17.8 [+16.0, +19.6]*

*Significant at $\alpha=0.05$ (two-sided paired bootstrap; both survive Bonferroni correction, $\alpha/4=0.0125$). p -values: news popularity $p=0.063$; news topic $p=0.081$; Reddit popularity $p=0.008$; Reddit topic $p<0.001$.

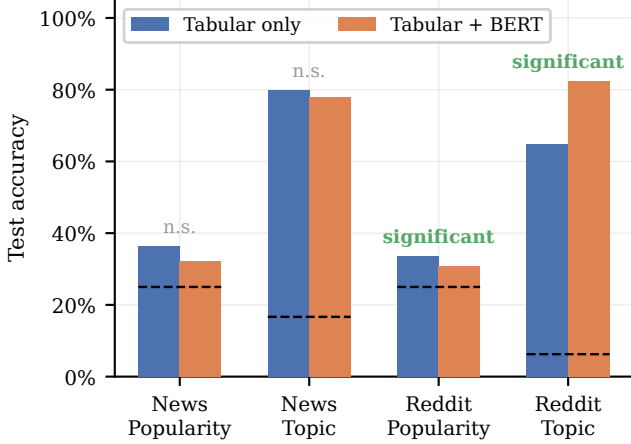


Fig. 2. Accuracy by task/domain/configuration against chance level (dashed). BERT’s contribution is large and significant only for Reddit topic classification.

(embedding + tabular fitting + attribution, all eight cells) completes in under an hour on this hardware; no distributed or multi-GPU setup is used or required. All code, exact package versions, and the fixed seed are in the released repository at <https://github.com/Ayush121511/OnlineNewsPopularity>.

V. RESULTS

A. Significance and meaning of headline results

Table IV and Figs. 2–3 show the entire 2×2 result matrix. The effect of BERT is statistically significant only in exactly two out of the four cells, oppositely: a large and very significant *positive* effect on Reddit topic (subreddit) classification, and a small yet significant *negative* effect on Reddit popularity. For both news-domain cells, BERT has a point estimate that is negative in direction, though not statistically significant to report. Notably, News/Topic is the domain’s designated positive-control task (§III) — its label is a direct function of article content — yet BERT’s effect there is statistically indistinguishable from zero, no different from News/Popularity, where no such expectation holds. This is not a failure of the control to detect a real effect: §V-B shows that LDA topic-model features already capture most of the same signal, leaving little room for BERT to add anything beyond what the tabular block already encodes.

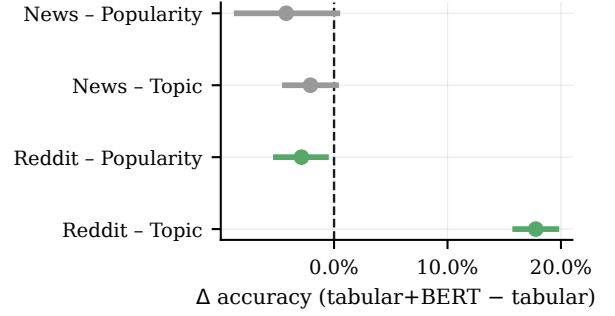


Fig. 3. Paired-bootstrap 95% confidence intervals on the accuracy delta. Only the two Reddit cells exclude zero.

B. Where does the signal live? Group importance

The per-feature-normalized SHAP group importance is illustrated in Figs. 4 and 5 for each cell of the tabular+BERT model. Across both domains, three consistent patterns emerge:

- **Popularity is structural.** In both the news and Reddit datasets, the structural group (link/keyword statistics in news and post-structure features in Reddit) ranks highest according to the per-feature SHAP share, even when BERT is included. It accounts for 39.8% of the per-feature SHAP share in news and 76.0% in Reddit, while BERT accounts for 6.6% and 13.5%, respectively. This is despite BERT contributing 83–95% of the raw SHAP sum because it consists of 768 features (Table V). This illustrates the type of feature-count bias that per-feature SHAP is designed to address.
- **News topic is already represented by LDA.** The LDA group consists of five membership scores derived from LDA models using the same article text that is also embedded by BERT. This group dominates news topic classification, accounting for 82.5% of the per-feature SHAP share in the tabular+BERT model, whereas BERT contributes only 1.2%. A reasonable null hypothesis is therefore that LDA membership scores provide a task-specific representation of the same underlying signal captured by BERT.
- **Reddit topic is where BERT succeeds.** Subreddit is the only label in this study that depends directly on text, vocabulary, and wording—for example, a post in *r/Jokes* differs from a post in *r/science*—without additional feature engineering such as LDA. BERT accounts for 72.5% of the raw SHAP sum and 68.0% of the permutation accuracy drop, leading on both (Table VI), although not on the per-feature SHAP metric (1.6%, Table V). This is the only cell where BERT clearly leads on a metric that is not inflated by its dimensionality.

C. Cross-method robustness: SHAP vs. permutation importance

Table VI and Fig. 6 compare BERT’s importance share under SHAP against its share of accuracy drop under group

TABLE V
RAW VS. PER-FEATURE-NORMALIZED SHAP SHARE FOR THE BERT
GROUP, TABULAR+BERT MODELS (ILLUSTRATES THE DIMENSIONALITY
ARTIFACT)

Domain	Task	BERT, raw % SHAP	BERT, per-feat. %
News	Popularity	83.1%	6.6%
News	Topic	57.6%	1.2%
Reddit	Popularity	95.5%	13.5%
Reddit	Topic	72.5%	1.6% ^a

^aPer-feature share is low even here because the comparison groups (*structure*, *source*) have very few features (3–6) that each carry a large individual share; BERT nonetheless holds the largest permutation accuracy-drop share in this cell (68.0%, Table VI); its largest-*raw*-share position holds in every cell and reflects its dimensionality.

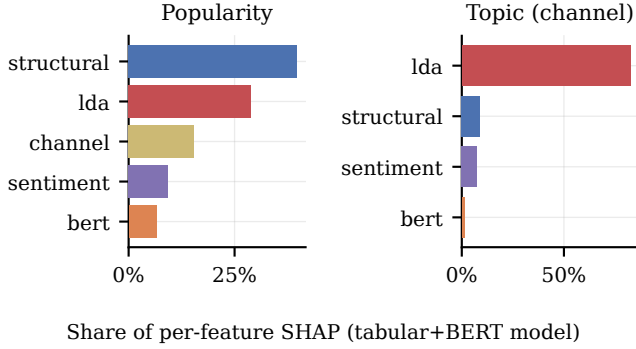


Fig. 4. News domain: per-feature-normalized SHAP group importance, tabular+BERT model. Structural features dominate popularity; LDA dominates topic. BERT is not the top group in either task.

permutation, computed independently on the same fitted models. The two methods agree on the dominant structured feature group in all four cells (structural for news popularity, LDA for news topic, structure for both Reddit tasks), and both place BERT last in the two news cells. They diverge on Reddit: for topic classification, permutation importance ranks BERT first (68.0% of the drop), in line with its raw SHAP share (72.5%) but not its per-feature share (1.6%); for popularity, BERT’s permutation share (48.4%) is tied with structure (48.2%) even though adding BERT lowers held-out accuracy, a pattern consistent with the 768-dimensional block crowding the few structured features out of the Random Forest’s per-split feature subsets. The agreement on the dominant structured group is evidence that our central finding is not an artifact of TreeExplainer specifically.

VI. DISCUSSION

The above four cells thus suggest an important constraint on the application of BERT rather than an absolute yes or no. Specifically, the distinguishing factor is not whether the domain is news or Reddit nor whether the prediction target is popularity or topic. Instead, it lies in the nature of the target: is the target determined more by the semantic meaning of content or rather by the behavioral and structural features?

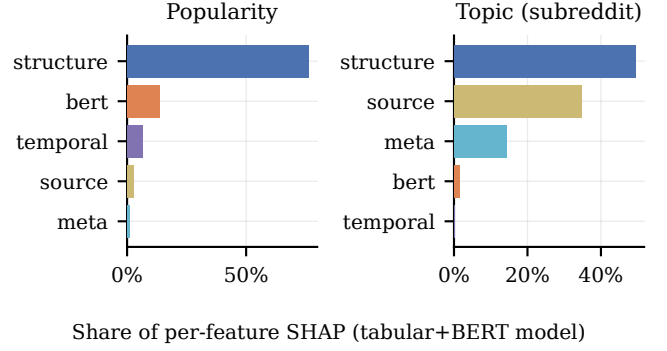


Fig. 5. Reddit domain: per-feature-normalized SHAP group importance, tabular+BERT model. Structure dominates both tasks; for topic (subreddit), BERT ranks second-to-last on this metric (1.6%) despite leading on permutation importance (Table VI) and raw SHAP sum, illustrating the metric divergence discussed in §V-C.

TABLE VI
BERT GROUP’S IMPORTANCE SHARE, TWO INDEPENDENT METHODS

Domain	Task	SHAP (per-feat. %)	Perm. (% of drop)
News	Popularity	6.6%	0.0% ^a
News	Topic	1.2%	2.2%
Reddit	Popularity	13.5%	48.4%
Reddit	Topic	1.6%	68.0%

^aPermuting BERT columns in the news-popularity model *improved* held-out accuracy on average (negative drop, clipped to 0% of share) — i.e. BERT was actively adding noise, not merely failing to add signal.

- *Popularity is structural in both domains.* In the news domain, popularity depends mainly on contextual features such as links and keywords. In Reddit, structure and timing of posts matter. Such features only weakly correlate with text meaning, and hence the inclusion of frozen text embeddings does not contribute any meaningful additional information and can actually add noise. In Reddit, this noise is significant enough to decrease the accuracy, despite a vastly larger training dataset (15,998 vs. 3,437 rows), which in theory should give more data for fitting a 768-dimensional feature block.
- *Topic classification of the news channel is a case of feature redundancy.* In this task, an engineered feature, namely LDA topic membership, already captures most of the information required for the prediction. Hence, BERT adds information that overlaps largely with the tabular feature space.
- The only task in this matrix which lacks any engineered proxy for topical/lexical content, with a label which is nearly definitionally a function of word choices, is the Reddit subreddit classification problem. And here is precisely where *BERT earns its dimensionality* by an order of magnitude and a statistically significant margin.

The findings thus suggest a heuristic by which the need for a text embedding layer can be decided based on the type of label involved rather than the domain or the task name. The key is whether the label in question is essentially dependent

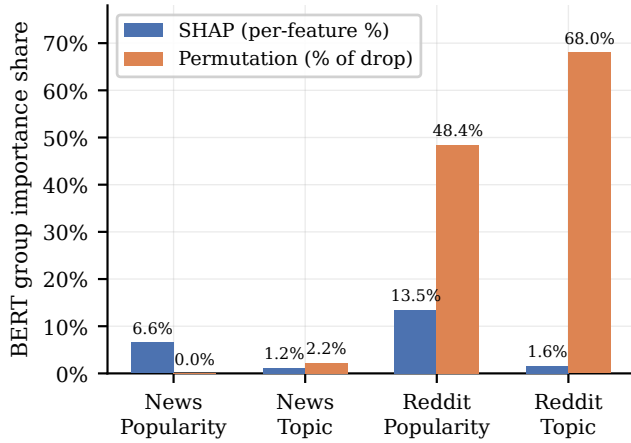


Fig. 6. BERT’s importance share under SHAP (per-feature-normalized) and permutation importance (% of accuracy drop), computed independently. Both methods show BERT contributing little in the news cells; they diverge on Reddit, where BERT’s permutation share far exceeds its per-feature SHAP share (§V-C).

on the textual/semantic component, or it is dependent on structural/behavioral component that happens to coincide with the text. A proxy feature derived from the text can help identify such a label; when the label can already be predicted reasonably well by an easy-to-understand text-derived feature, this means that the label is a textual label, and thus the added value of a frozen embedding layer is low. But when there is no such feature but the label depends on the choice of the words and the way they are phrased, then this is precisely the place where the frozen embedding helps — as shown by our Reddit topic experiment.

VII. THREATS TO VALIDITY

Only one family of models. All results come from Random Forest only. We make no claim about generality to gradient-boosted trees [20] or a tuned neural fusion model; the permutation importance cross-check in §IV-C argues against a TreeExplainer-specific effect but not a Random-Forest-specific one.

Frozen, not fine-tuned, embeddings. A fine-tuned DistilBERT could possibly fix the issue with the negative result; this is a deliberate design choice to isolate the “off-the-shelf embeddings as a tabular feature” case, which is widespread, not the case of what transformers could achieve with tuning.

Single fusion strategy. For the case under examination, only one approach to combining the BERT embedding and the tabular model has been used: simple concatenation into one vector. Although there are ways of fusion that are much more expressive — including late fusion with a gating weight, cross-modal attention [9], and other schemes described in [10] — it is theoretically possible that some signal is lost by the concatenation approach, particularly in the two news-domain cells where null results have been obtained. Therefore, our findings apply exclusively to this setting (frozen DistilBERT, Random Forest, concatenation).

Smaller sample size in news domain. The sample size (3,437 articles) is rather modest; two of the confidence intervals concerning the news domain are too broad ($p = 0.06$ – 0.08); a larger scrape would easily tip the results in either direction; we state this frankly as a null result, not as a confirmation that BERT does not work on news domain.

Multiple comparisons. Four tests of significance are run without multiple comparison correction; both significant results ($p = 0.008$, $p < 0.001$) still hold even after the rather conservative Bonferroni correction ($\alpha = 0.05/4 = 0.0125$).

VIII. CONCLUSION

The study runs a controlled, statistically grounded test of whether a frozen BERT embedding adds predictive signal beyond structured and metadata features, across two domains, two tasks, and two independent attribution methods, backed by paired-bootstrap significance testing rather than a single point estimate of accuracy. The result we arrive at gives a clear boundary condition for the decision on using BERT embeddings: BERT delivers a large, significant gain only where the label is intrinsically lexical and no engineered feature already proxies for it (17.8 points, $p < 0.001$, Reddit subreddit classification); everywhere else in the grid, it adds nothing, and in one case actively hurts accuracy. The dimensionality cost of a text embedding is worth paying under exactly one condition: the label is topical or lexical in nature, and no engineered feature already captures that signal.

ACKNOWLEDGMENT

The author thanks the Yardi School of Artificial Intelligence, IIT Delhi, for the computing and academic environment supporting this work. Generative AI (Claude, Anthropic) was used, under the author’s direction, to assist with drafting and language refinement of the manuscript text and with implementing the figure-generation code underlying the reported results. All experimental design choices, results, and conclusions were reviewed and verified by the author.

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